

ChE 381/382: Unit Operations Laboratory

Heat Transfer to a Fluid in a Stirred Tank

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Introduction

Stirred tanks are widely used in the chemical process industry. They come in a great variety of sizes and materials of construction. A typical system consists of a stirred tank assumed to be perfectly mixed or at constant bulk properties, a steam jacket to provide heating, and an internal cooling coil. Baffles are often used to enhance mixing. Additional vessel cooling can be further achieved by a recirculation circuit in which the process fluid is passed through an external heat exchanger. Examples of use include simple mixing of solutions, chemical reactions (CSTR), and leaching operations [1].

Heat transfer in these systems is often an important factor. It might be necessary to add heat to the fluid to avoid phase changes, or to cool the system in the case of an exothermic reaction. In many cases, proper heat transfer design is crucial for successful use of a stirred tank.

A stirred tank system will be used to study a range of heat transfer processes. The focus will be on the measurement of the heat transfer coefficients associated with the heat transfer across the vessel wall. Parametric studies will include the dependence of the heat transfer coefficients on the impeller speed and the use of mechanical baffles. The experimentally measured heat transfer coefficients can also be compared to values reported in literature.

Theory [1-5]

A brief synopsis of the theory for heat transfer in a stirred tank will be given here. For more complete details on heat transfer, refer to a chemical engineering handbook, heat transfer or transport phenomena textbook (see references for examples).

A microscopic view of heat transfer from one fluid to another across a solid wall reveals a number of processes in series. In Figure 1, it is assumed that heat transfers from right to left. Often, a layer of scale (mineral deposits) forms on one or both sides of the wall. Heat must transfer across the hot fluid-scale interface, be conducted across the scale and the solid wall, and finally across the cooler side of the wall-scale-fluid interface. At steady state, the heat transferred, Q , is the same at any point in the x -direction.

In the case of the stirred tank, the heat transfer process consists of three basic steps:

- Heat transfer from the condensing steam in the steam jacket to the scale-wall
- Heat transfer across the solid wall.
- Heat transfer from the wall-scale of the stirred tank to the vessel fluid (water).

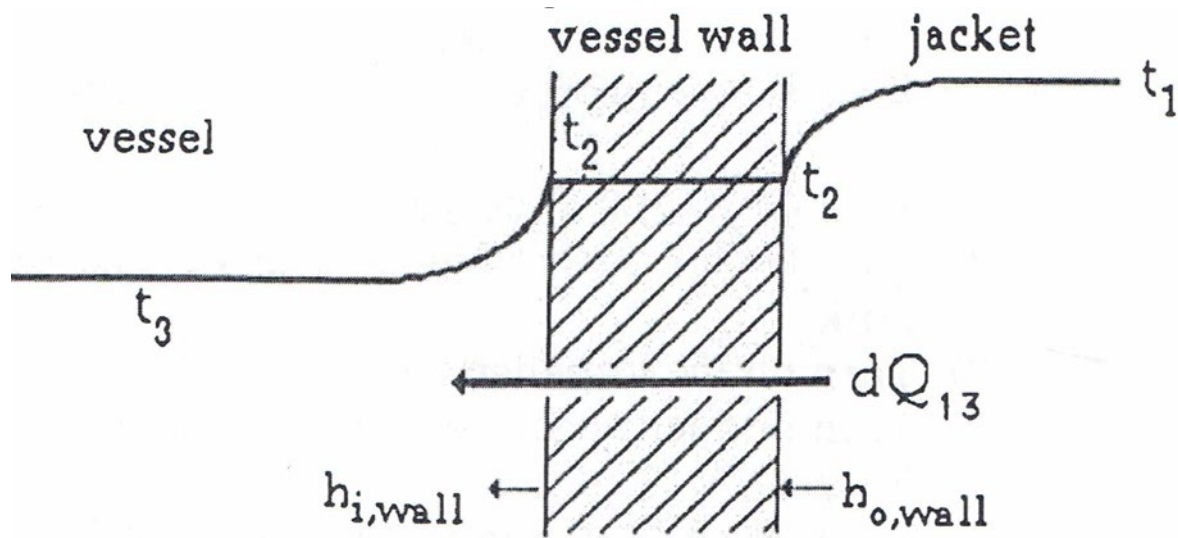


Figure 1. Heat transfer across a vessel wall.

In general, heat transport can be expressed by Fourier's Law:

$$Q = kA \frac{dt}{dx} \quad (1)$$

For the purposes of this experiment, calculations for the heat transfer coefficients will be based on these assumptions:

- No scale on either side of the solid wall
- Negligible resistance to heat transfer by the wall. In other words, the temperature gradient across the solid wall is small. This occurs if the thermal conductivity of the solid is high, as would be expected for a metal wall.
- Wall thickness is small compared to the tank diameter such that the inner wall area $\approx$ outer wall area.

Each interface and solid phase present a different resistance to heat transfer such that the temperature profile at steady state is established as shown in Figure 1. Q can be expressed for the two liquid-solid interfaces in the stirred tank as:

$$Q_{13} = h_o A_o (t_1 - t_2) = h_i A_i (t_2 - t_3) \quad (2)$$

Where h_i is the film heat transfer coefficient at the inner surface with area A_i , h_o is the heat transfer coefficient at the outer surface with area A_o , t_1 is the temperature of the hot fluid (steam), t_2 is the temperature of the inner and outer wall, and t_3 is the temperature of the cooler fluid (stirred tank water).

Note that the heat flux, Q_{13} , through each of the individual resistances must be the same and equal to the total heat flux across the interface since at steady state, there can be no accumulation of heat and the heat flux across the interface is a constant.

The above equations for heat transfer across each individual resistance in terms of the local interface temperatures can be combined and expressed in terms of the temperature difference ($t_1 - t_3$), across the composite interface and an overall heat transfer coefficient, U_i . If the area A_i is considered from the above assumptions, equation (1) becomes:

$$Q_{13} = k_i A_i \left(\frac{t_1 - t_3}{x} \right) = U_i A_i (t_1 - t_3) \quad (3)$$

In addition, based on the assumptions made above, a simple expression for the overall heat transfer coefficient, U_i in terms of h_i and h_o can be written:

$$U_i = \frac{1}{\frac{1}{h_i} + \frac{1}{h_o}} \quad (4)$$

Measurement of Heat Transfer Rate: Energy Balances

Figure 2 shows the mass and energy flow in a stirred tank with a steam jacket, steam condenser, internal cooling coil, and external heat exchanger.

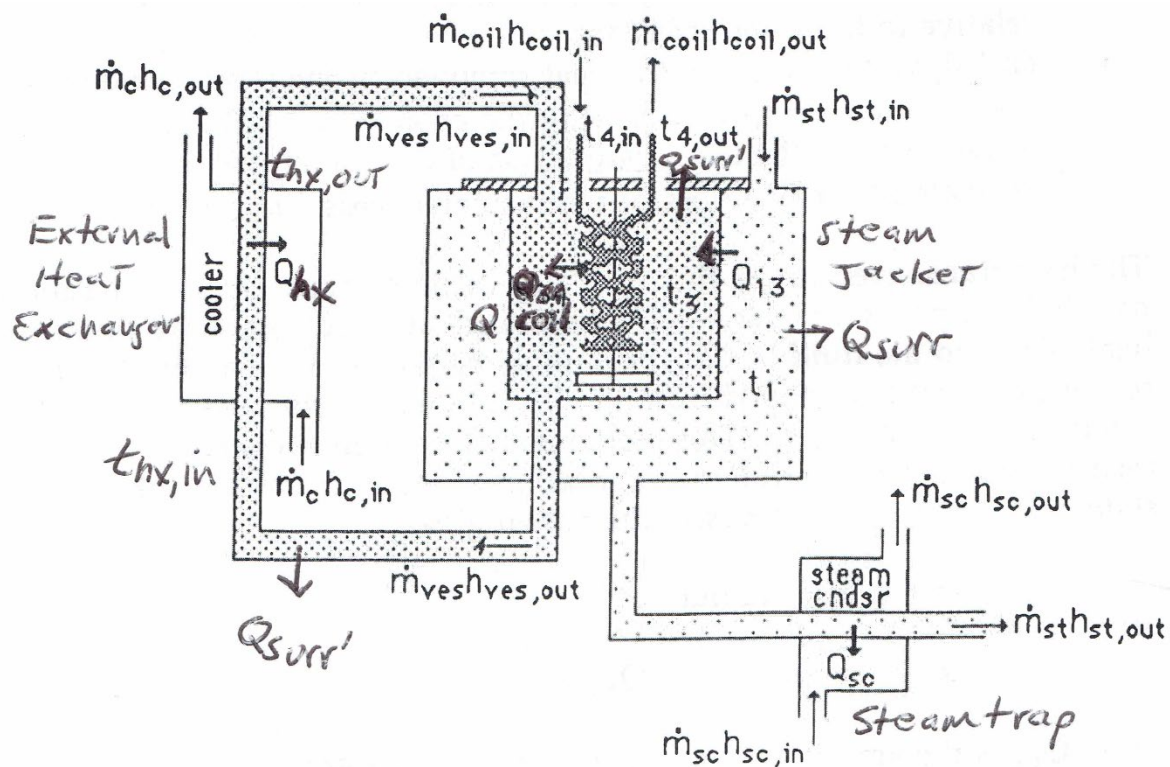


Figure 2. Schematic of mass and energy flow in a stirred tank.

Method 1: Steam Energy Balance

The heat transferred to the vessel from the steam, Q_{13} , can be determined from an energy balance on a control volume around the steam jacket and steam condenser:

$$Q_{13} = \dot{m}_{st} (h_{st,in} - h_{st,out}) - Q_{sc} - Q_{surr} \quad (5)$$

Where $\dot{m}_{st}$ is the mass flow of the steam, $h_{st,in}$ is the enthalpy of the entering steam, $h_{st,out}$ is the enthalpy of the steam condensate, Q_{sc} is the heat removed in the steam condenser, and Q_{surr} is an estimate of the heat loss of the insulated vessel to the surroundings.

Method 2: Stirred Tank Fluid Energy Balance

The heat transferred to the fluid can also be determined by consideration of a heat balance around the stirred tank vessel since at steady state the amount of heat added to the fluid in the vessel must equal the amount of heat removed. Heat is removed from the fluid by either the cooling coils, the external heat exchanger, and from heat loss to the surroundings. A steady state heat balance, heat in = heat out gives the following equation:

$$Q_{13} = Q_{hx} + Q_{coil} + Q_{surr'} \quad (6)$$

Where Q_{hx} is the amount of heat removed from the recirculation water in the external heat exchanger, Q_{coil} is the amount of heat removed by the cooling coils, and $Q_{surr'}$ is the heat loss from the recirculation pipes and water evaporation from the vessel. If negligible energy loss along the recirculation piping or from evaporation from the tank is assumed, then $Q_{surr'}$ is zero. Q_{hx} can be determined from the flowrate and temperature drop of the fluid recirculated through the heat exchanger:

$$Q_{hx} = \dot{m}_{ves} C_p (t_3 - t_{hx,out}) \quad (7)$$

Where $\dot{m}_{ves}$ is the amount of fluid recirculated through the external heat exchanger, C_p is the heat capacity of the fluid, t_3 is the temperature of the fluid entering the heat exchanger (this temperature should be the same as the vessel temperature if the fluid is well mixed), and $t_{hx,out}$ is the temperature of the fluid leaving the heat exchanger and returned to the tank.

The heat removed by the cooling coils can be determined from the flowrate of the water in the coils and the temperature rise of the water as it passes through the coils:

$$Q_{coil} = \dot{m}_{coil} C_p (t_{4,out} - t_{4,in}) \quad (8)$$

Where $\dot{m}_{coil}$ is the flowrate of water through the cooling coils, C_p is the heat capacity of the cooling water, and $t_{4,out} - t_{4,in}$ is the increase in temperature of the cooling water in the coils.

If all heat flows are correctly accounted for, Q_{13} determined by either control volume heat balance should be the same. However, Method 2 is preferred for this experiment.

Heat Transfer Coefficient Correlations

The values of heat transfer coefficients are functions of the fluid flow field and the molecular transport properties of the fluid. In many instances, such as a stirred tank, the flow pattern is too complex to be analyzed analytically. As a result, heat transfer rates must be determined experimentally. However, experimental data, theoretical analysis, and dimensionless analysis have shown that the heat transfer coefficient between a fluid and a surface can frequently be correlated in terms of a relatively small number of dimensionless groups. For convective flow, usually the Reynolds number (Re), Prandtl number (Pr) and geometric factors are needed to correlate heat transfer data over a broad range of conditions. Thus, heat transfer coefficients are often reported as functions of these dimensionless groups. Correlations have been formulated for design purposes for many standard cases. These can be found in engineering handbooks such as Perry's [3].

Experimental data for h_i , the inner wall heat transfer coefficient, for an unbaffled stirred tank containing a Newtonian fluid have been summarized by the following empirical correlation:

$$Nu_T = .36 * Re_{imp}^{\frac{2}{3}} * Pr^{\frac{1}{3}} \quad (9)$$

Where:

$$Nu_T = \frac{h_i D_T}{k} = \text{tank Nusselt number} \quad (10)$$

$$\frac{D_I^2 n \rho}{\mu} \equiv Re_{imp} = \text{impeller Reynolds number} \quad (11)$$

$$\frac{C_p \mu}{k} \equiv Pr = \text{fluid Prandtl number} \quad (12)$$

D_T = tank diameter

D_I = impeller diameter

μ = viscosity of the fluid

ρ = density of the fluid

k = thermal conductivity of the fluid

C_p = specific heat of the fluid

n = speed of rotation of the impeller (revolutions per unit time)

Since these groups are dimensionless, any consistent set of units may be used for their evaluation. Note that Reynolds, Prandtl, and Nusselt numbers are standard dimensionless groups, which usually appear in correlations of heat transfer coefficients. However, the characteristic length scales with respect to which these groups are defined will depend on the

geometric characteristics of the system. For example, in this correlation the Reynolds number is defined in terms of the impeller diameter and the impeller tip speed. The fluid properties used to evaluate these groups correspond to that fluid through which heat transport is taking place. It is important that the length scales and fluid properties be clearly defined in any correlation and care should be used that these groups are correctly evaluated in a specific correlation.

The heat transfer coefficient on the outside of the vessel wall, h_o , where film condensation from the steam jacket occurs, can be estimated by the empirical correlation:

$$\frac{h_o L_T}{k} = 0.925 \left(\frac{L^3 \rho^2 g}{\mu \Gamma} \right)^{\frac{1}{3}} \quad (13)$$

Which is valid for: $Re_{film} = \left(\frac{4 \Gamma}{\mu} \right) > 2100$,

where:

$$\Gamma = \frac{\dot{m}_{st}}{\pi D_{T,o}}$$

And:

$\dot{m}_{st}$ = mass rate of steam condensation

$D_{T,o}$ = outer diameter of the vessel wall

L_T = vertical length of the tank

μ = viscosity of the condensed steam

ρ = density of the condensed steam

g = acceleration due to gravity

No general correlations exist for the case of a stirred tank with baffles in a jacketed tank. The presence of baffles deters the formation of a vortex in the tank, the flow profile has more of a radial component, and the fluid is better mixed. Also, baffles permit the use of higher speeds. The net effect is that in a fully baffled vessel, the heat transfer can be many times higher than for an unbaffled vessel.

Experiment:

A. Description of Experiment

A schematic of the jacketed, stirred vessel with recirculation stream and internal cooling coil is shown in Figure 3. Steam enters the steam jacket and is controlled with an inlet valve. The condensed steam exits through a steam condenser. Cooling water enters the cooling coil through a rotameter which indicates the water flowrate. Analog dial thermometers are provided at the inlet and outlet of the cooling coil. Water from the tank is continuously withdrawn from the bottom of the vessel and recirculated via a pump through the shell side of an external heat exchanger where it is cooled by cold water flowing through the tube bundle. The cooled tank water passes through a rotameter and is returned to the vessel. The temperature of the recirculated tank water is measured both at the inlet and outlet of the heat exchanger. The water temperature inside the tank is also indicated by a thermometer located on the top lid of the vessel. The impeller speed is measured by a tachometer connected to the apparatus; impeller speed is varied with a variable power supply (Variac). Tap water is supplied through a hose connected to a manifold in the back of the apparatus.

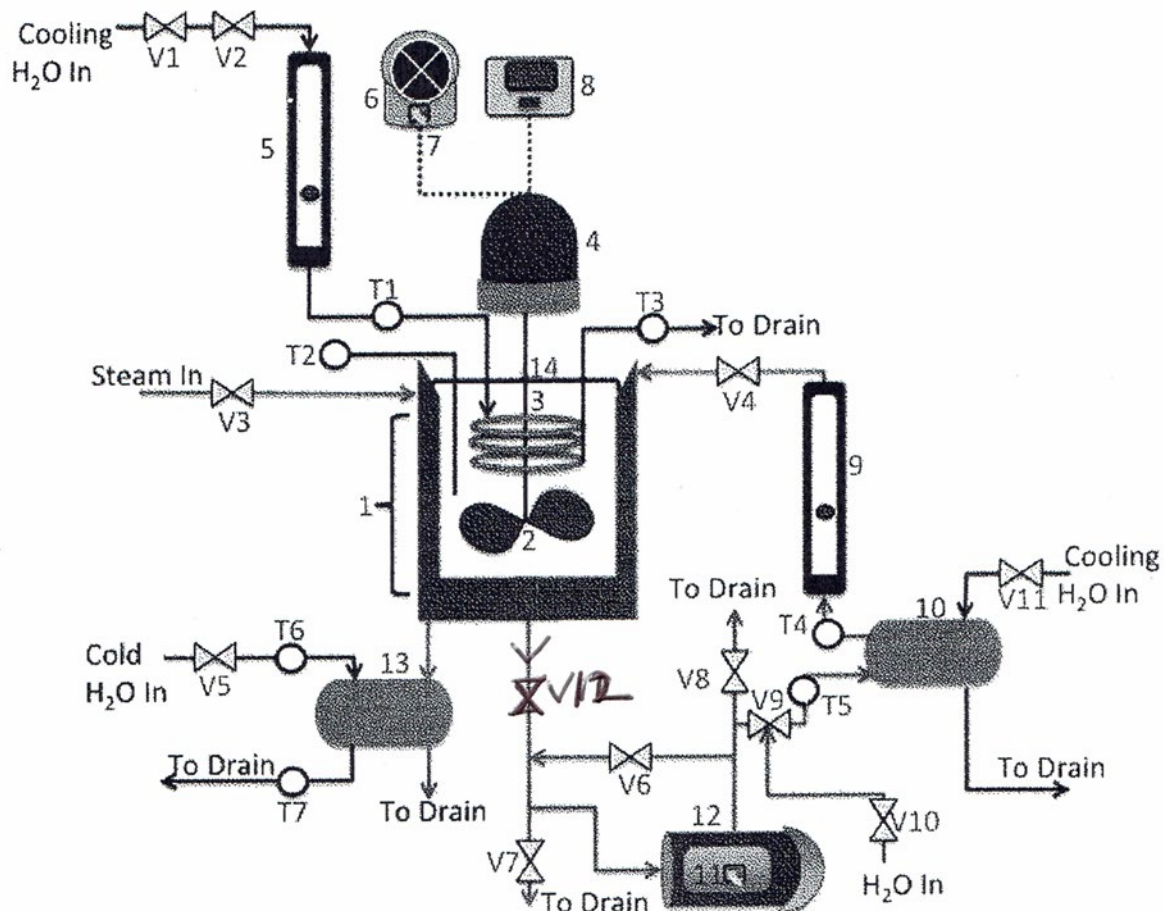


Figure 3. Schematic of the stirred tank apparatus

B. Experimental Procedure

Start Up Procedure

- Fill the tank with water to a level about 2-inches from the top of the tank. Be careful not to overfill the tank. Put the lid on top of the vessel.
- Turn on the impeller and adjust the speed to ~ 300 RPM. The reading will fluctuate, this is normal. Note the direction of the impeller rotation (clockwise or counterclockwise).
- Turn on the cooling water to the cooling coils (V1 and V2); open the control valve to obtain 50% flowrate in the rotameter.
- Turn on the cooling water to the tube side of the external heat exchanger (V11) and adjust the flowmeter to about 0.5 LPM.
- Open the steam condenser valve (V5) to allow cooling water flow through the steam condenser (steam trap).
- Priming the pump: Make sure the outlet valve (V9) in the discharge line of the pump is fully closed and the suction valve (V12) and bypass valve (V6) are fully open (why?). Switch on the pump and allow it to run for a few seconds in bypass mode to remove any air in the pump head and piping.
- Switch to Recirculation flow: Open valve V4. Open the discharge valve (V9) slowly to 100% flow and close the bypass valve (V6). Suction valve V12 stays open. Let the water flow at 100% flowrate to purge air in the line, then reduce the flow to about 50%. Verify that the circulating fluid is being fed back to the tank.

Experiment 1: Unbaffled Stirred Tank

Devise a set of experiments to study the effect of impeller speed, flowrate of water through the heat exchanger and vessel temperature on the Q_{13} , heat transfer coefficient, h_i , and the overall heat transfer coefficient, U_i . The set of process variables (cooling coil flowrate, recirculation flowrate in external heat exchanger, steam rate) should be selected to keep the tank temperature around 65°C and not more than 85°C .

1. Open the steam valve (V3) slightly, about 1 to $1\frac{1}{2}$ turns, from the closed position to allow steam to enter the steam jacket and condense completely. Later calculations will be simplified if the steam flow rate is low enough so that all the steam condenses completely in the jacket. This will be evidenced by only a small temperature increase in the shell side fluid of the steam condenser from the two dial thermometers T6 and T7.
2. Set the impeller speed to the first value. Start making periodic measurements of all the temperatures and flowrates in order to determine when the temperatures have stabilized, and a steady state is reached. This may take some time as the temperature in the vessel increases and the inlet/outlet temperatures on the cooling coil and water recirculation line stabilize. Note: The cooling coil water flowrates or the recirculation flow in the external heat

exchanger may need to be adjusted if your vessel temperature does not stabilize between 65°C and 85°C.

3. After a steady state is reached, record the following data:
 - Flow rates of the cooling coil water and recirculation flow.
 - Temperatures from all the temperature gauges (T1 to T7).
 - The amount of steam condensate collected from the steam condenser over a time range using a graduated cylinder. This will be used to calculate the steam flowrate.
4. Repeat the above steps 1-3 for two more impeller speeds. Adjust the recirculation water flow rate to maintain the vessel temperature from above. Make sure the system has reached a steady state before recording the data for your calculations. Note: If you see a vortex in the middle of the fluid in the tank, lower the impeller speed.

Experiment 2: Baffled Stirred Tank

Conduct a series of experiments at the same conditions as in Experiment 1 except with the 4 baffles inserted into the vessel. **NOTE: These experiments can most easily be done in conjunction with those in the previous section.** For example, after recording all data at an impeller speed, turn the pump off, carefully lift the vessel cover plate off, and slip the baffles in place making sure the top flat section is seated over the Allen screw heads. Turn the pump back on. After a new steady state is reached, take all necessary measurements. Then remove the baffles and conduct the Experiment 1 trial for the next impeller speed and repeat.

C. Results and Discussion

The results below should be done for all trials in Experiment 1 and Experiment 2:

- Calculate Q_{13} using one of the energy balances.
- Calculate the overall heat transfer coefficient, U_i , using the value of Q_{13} .
- Calculate the outer heat transfer coefficient, h_o , using the correlation given.
- Calculate the heat transfer coefficient, h_i , from the experimental results using equation 4. Compare with the h_i obtained from the correlation (Equation 9).

The results can be summarized in tables as well in graphs of U_i , h_i , and h_o versus impeller speed, recirculation flow rate, for the baffled and unbaffled trials.

State trends in the heat transfer coefficients with impeller speed and any effects of the baffles. Does the change in flow pattern with the baffles increase the rate of heat transfer?

Compare the experimentally calculated heat transfer coefficient h_i , with those obtained from the correlation. Agreement may not be that good as the correlation given in

equation 9 is for a paddle stirrer whereas in this experiment an impeller is used. Explain the trends that you observe and if there is general agreement.

Overlay the baffled results with the unbaffled results. Is the heat transfer coefficient, h_i influenced by the baffles? Does the data show the same dependence on Re_{imp} for the baffled and unbaffled trials?

Further discussion:

Since no general correlations exist for the case of a stirred tank with baffles in a jacketed tank how would one obtain a new correlation for an impeller-stirred tank with baffles using the Nu , Pr and Re_{imp} ? How would you obtain new values for the exponents on Pr or Re_{imp} and the constant in the new correlation?

References

- [1] McCabe W.L., Smith J.C., Harriot P., "Unit Operations in Chemical Engineering" 7th Edition Section III Chapters 10-15, (2005)
- [2] R.B. Bird, W.E. Stewart, E.N. Lightfoot, Transport Phenomena, 2nd Edition, John Wiley and Sons, Inc., 2002.
- [3] R.H. Perry and D.W. Green, Eds., Perry's Chemical Engineering Handbook, 8th Edition, McGraw-Hill, 2008.
- [4] T.L. Bergman, F.P. Incropera, D.P. DeWitt, A.S. Lavine, Introduction to Heat Transfer, 6th Edition, Wiley & Sons, 2011.
- [5] A. M. Flynn, T. Akashige, L. Theodore, Kern's Process Heat Transfer, Scrivener Publishing LLC, 2019.